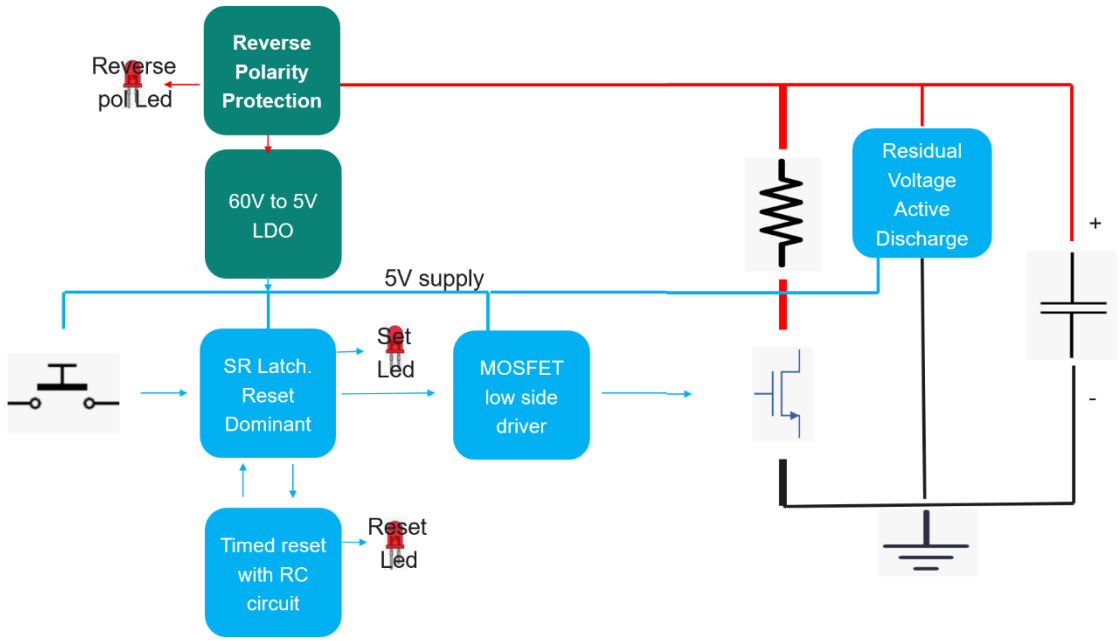
Practice Project – Discharger 60V 100mF

A.Schematic Design

1. Block Diagram

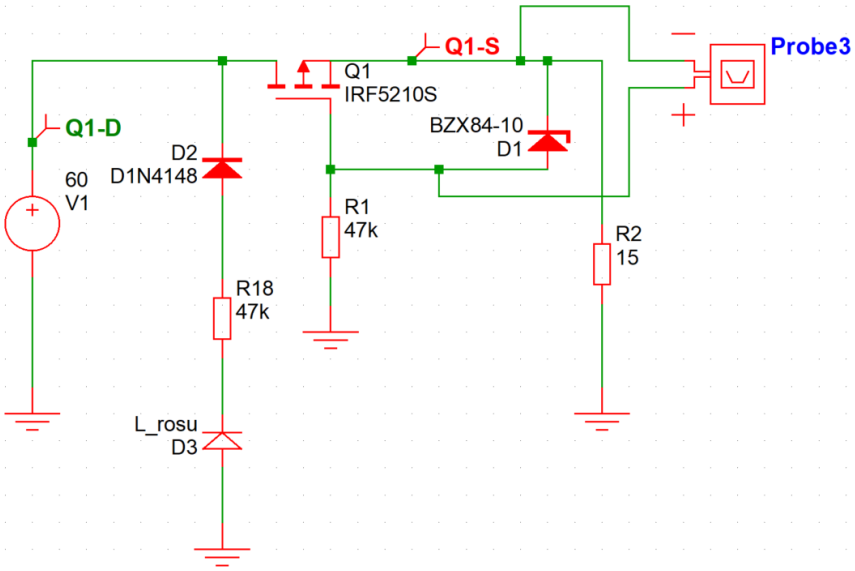


As you can see, my discharger system can be summarized by this block diagram. Each individual block performs a specific, straightforward function, and when combined, they create the entire system.

It is important to mention that the capacitor I want to discharge actually acts as the power source for the rest of the circuit – practically, it is the heart of the system. This self-powered design is exactly what makes the entire device fully portable.

Next, I will present and break down each individual block in a simple and easy-to-understand way. For every single block, I will walk you through both the simulation and its schematic implementation in Altium, exactly how I analyzed it myself.

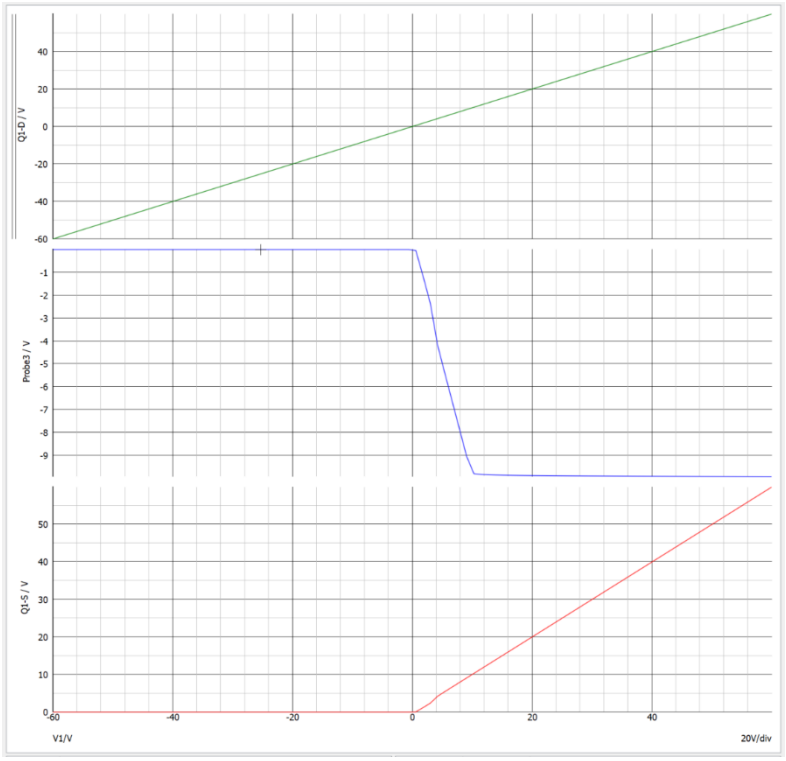
1.1. Reverse Polarity Protection Simulation & Schematic



The first functional block is this reverse polarity protection circuit, featuring a PMOS and Zener clamping on gate-source.

What is its main role? Well, if the capacitor is connected with reversed polarity, the PMOS isolates the rest of the circuit, and an LED turns on to signal that an error has occurred.

Meanwhile, the Zener diode clamps the maximum gate-to-source voltage, ensuring the transistor operates safely without being destroyed.

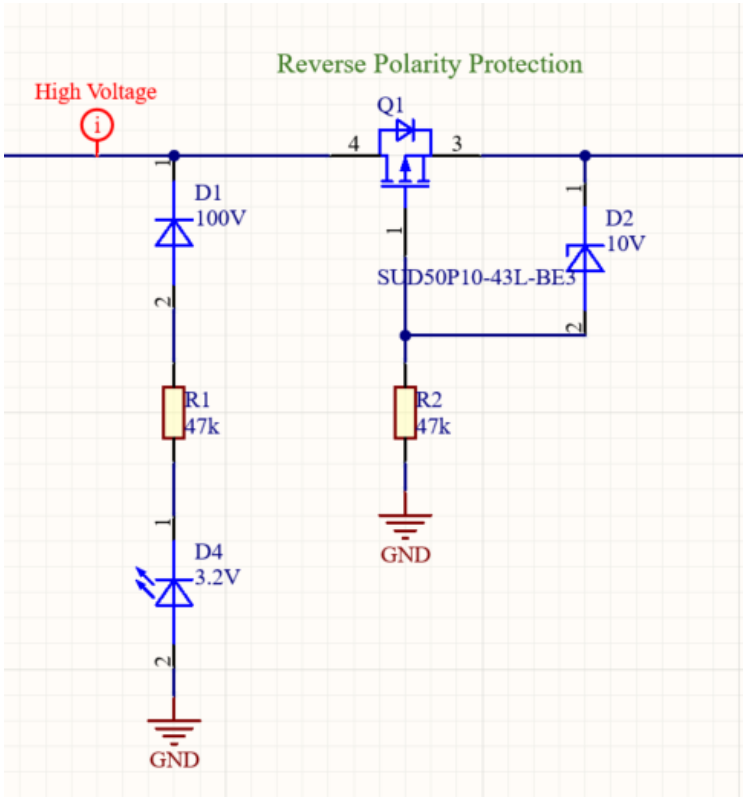


Input voltage

Gate-source

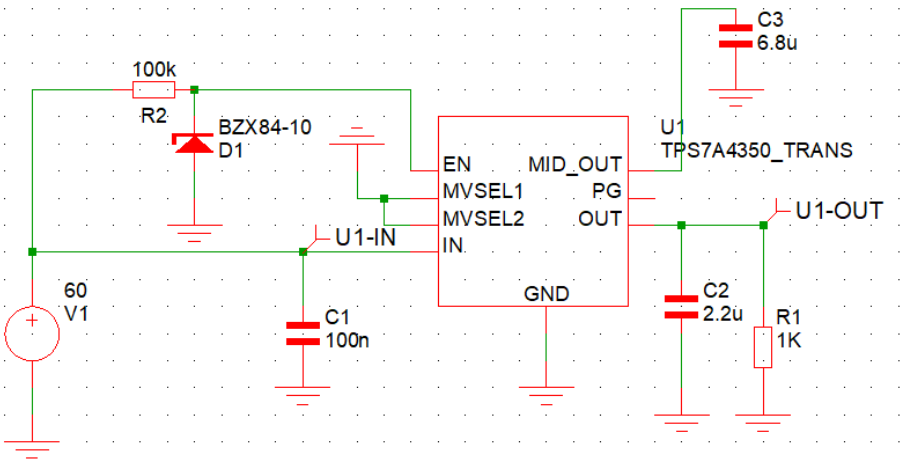
Output

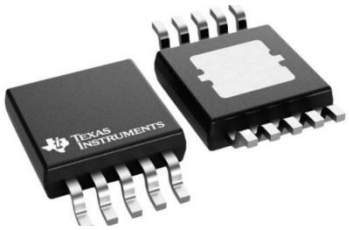
In this graph, I have validated the behavior I described earlier. You can clearly see the protection against negative input voltages, as well as the clamping at a safe level between the gate and the source of the transistor.



Here, we can see the Altium implementation of this block, detailing the protection circuit along with the LED signaling.

1.2. 60V to 5V LDO Simulation & Schematic





Why this LDO?

- Input voltage: 4 V to 85 V
- Operating junction temperature: –40°C to +125°C

Thermal calculation:

P=(Vin-Vout)*I=(60-5)*0.013=715mW

deltaT=P* RθJA=0.715*53.7=38 °C

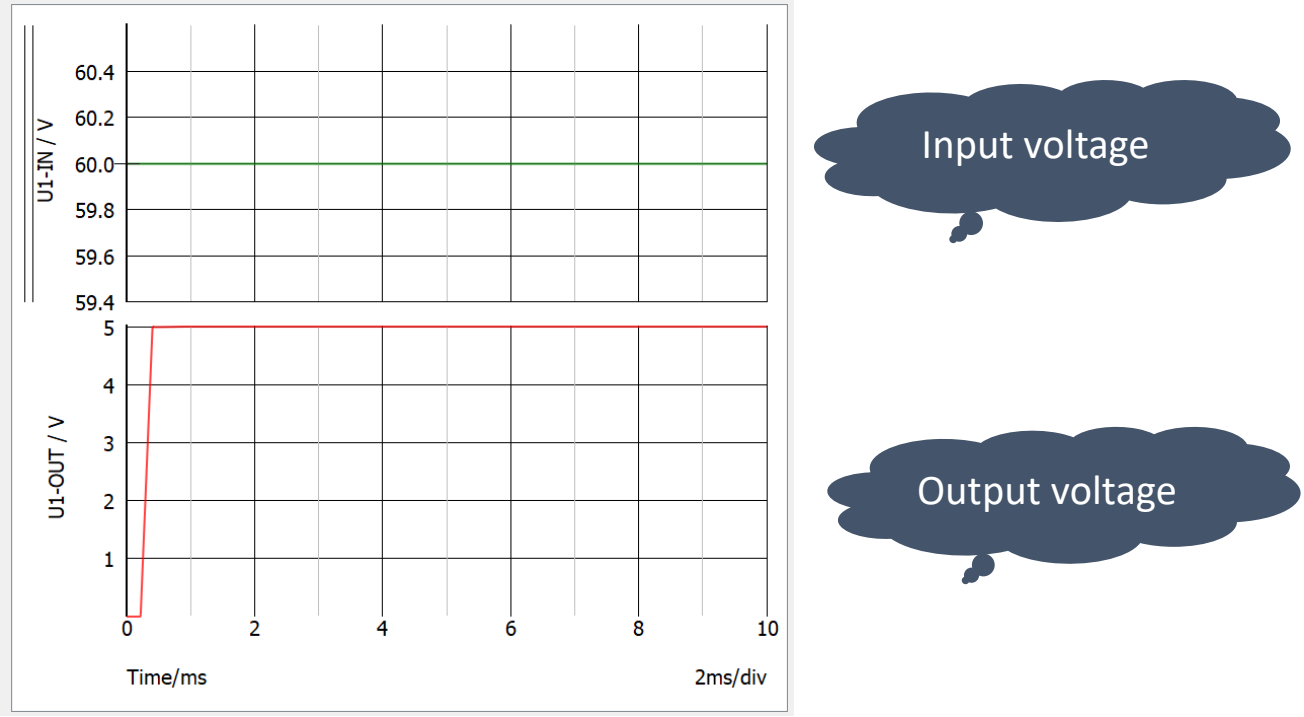
Tmax= Tamb + deltaT = 25 + 38 = 63 °C

- ✓ **63 °C < 125 °C**

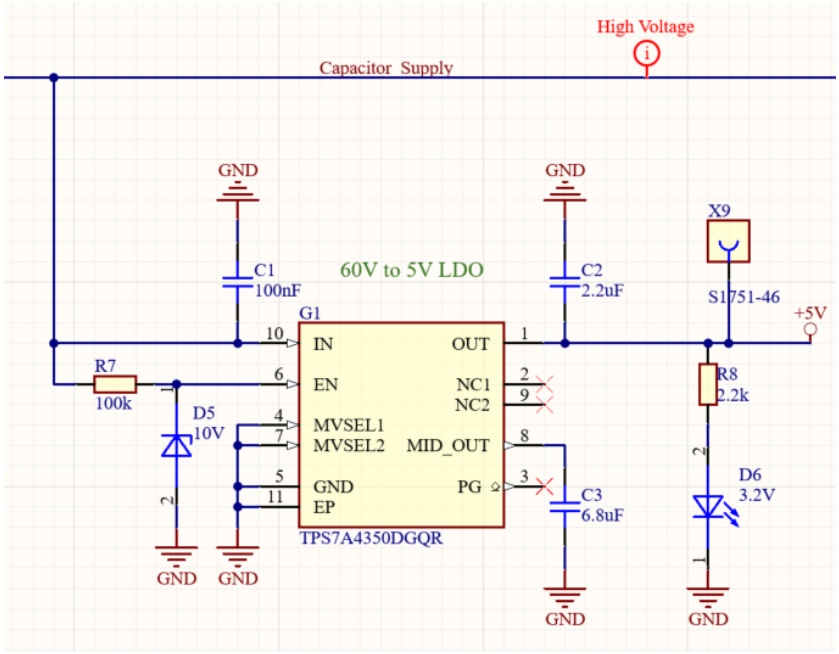
Moving forward, the next block acts as the bridge between the high voltage I want to discharge and the stabilized 5V that powers the control circuitry.

During my research, I looked for an LDO model that features a very wide input voltage range. Despite the large voltage drop between the input and output, which is converted into heat, the regulator operates within its safe operating parameters.

According to the simulations, the output current is 13mA. Based on the thermal calculations I performed by extracting the essential parameters from the datasheet, my component selection is fully validated.



The simulation graph confirms the functionality of this block, showing the 60V successfully converted into a stable 5V output.



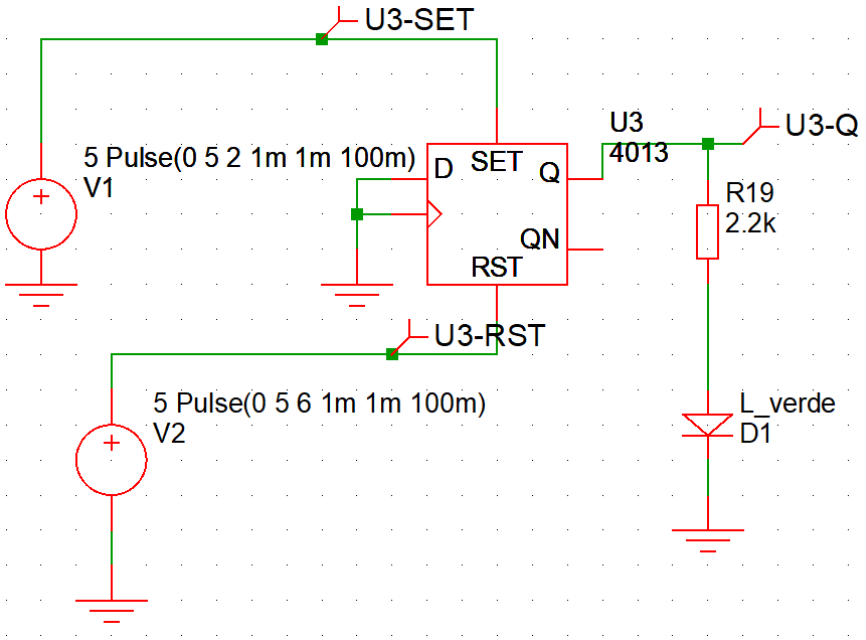
Here, we can see the schematic implementation in Altium, following the datasheet recommendations regarding capacitor selection and placement.

Parameter	LDO	DC-DC Converter
Efficiency	Low (decreases drastically with the Vin-Vout difference)	High (even at large differences)
Noise (EMI)	Very low, clean for analog logic	High, requires additional filtration
Complexity & Cost	Reduced (few external components)	High (requires inductors, capacitors)
Conversion Capability	Voltage reduction only	Step-Down, Step-Up or Inversion
Best case scenario	Powering noise-sensitive sensors and circuits, low power	High power, high current applications

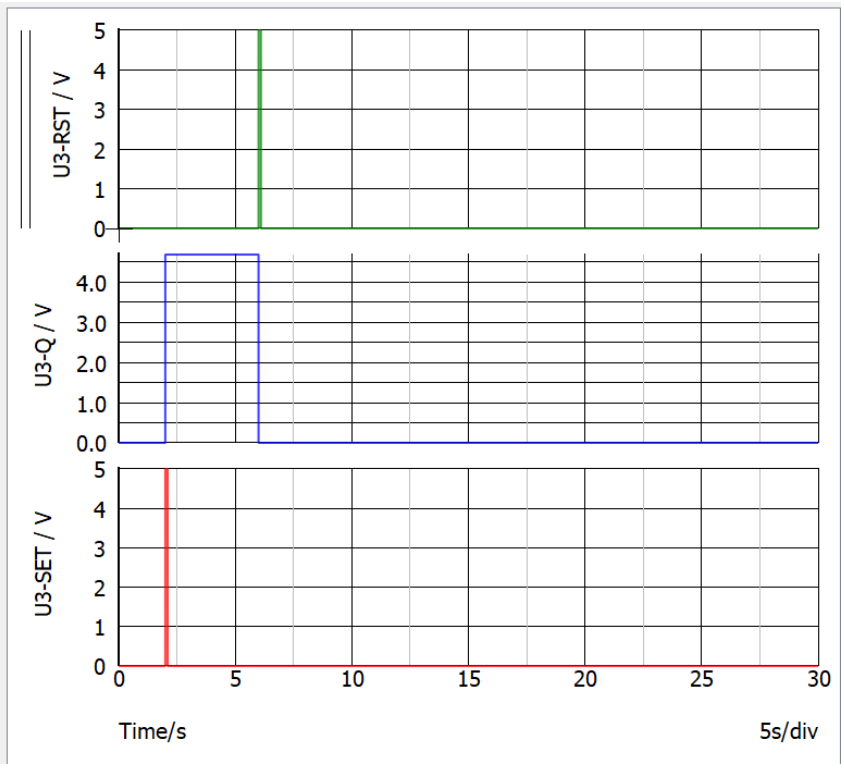
At this point, I faced my first major design decision. My motivation for choosing an LDO over a DC-DC converter was its ultra-low noise performance, which is essential for sensitive circuitry, as well as its low complexity, given my strict space constraints.

The trade-off, however, was the low efficiency resulting from the large difference between the input and output voltage.

1.3. SR Latch Simulation & Schematic



Moving into the digital circuitry domain with the SR Latch, the logic here is very straightforward.



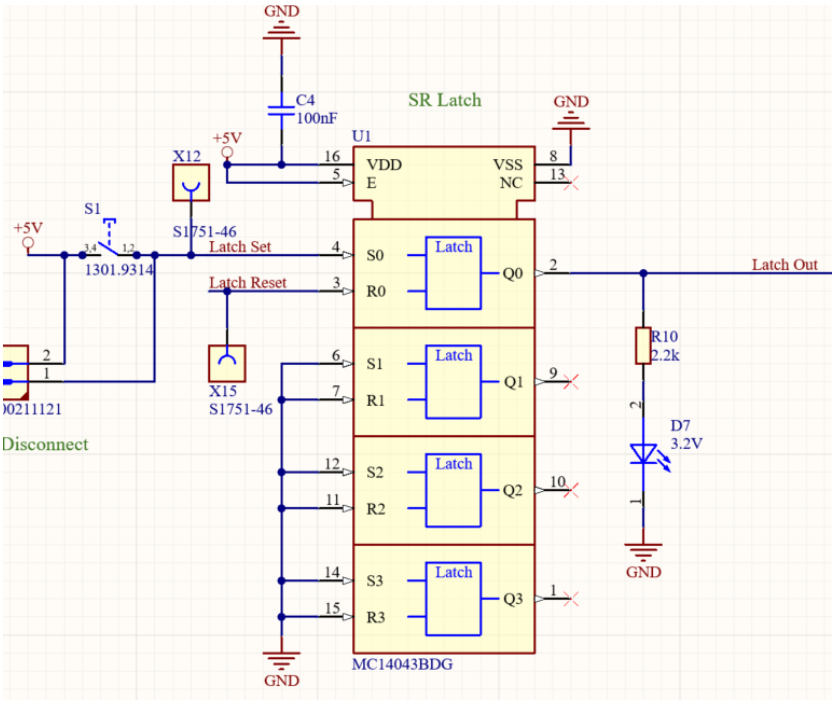
RESET Latch

Output Latch

SET Latch

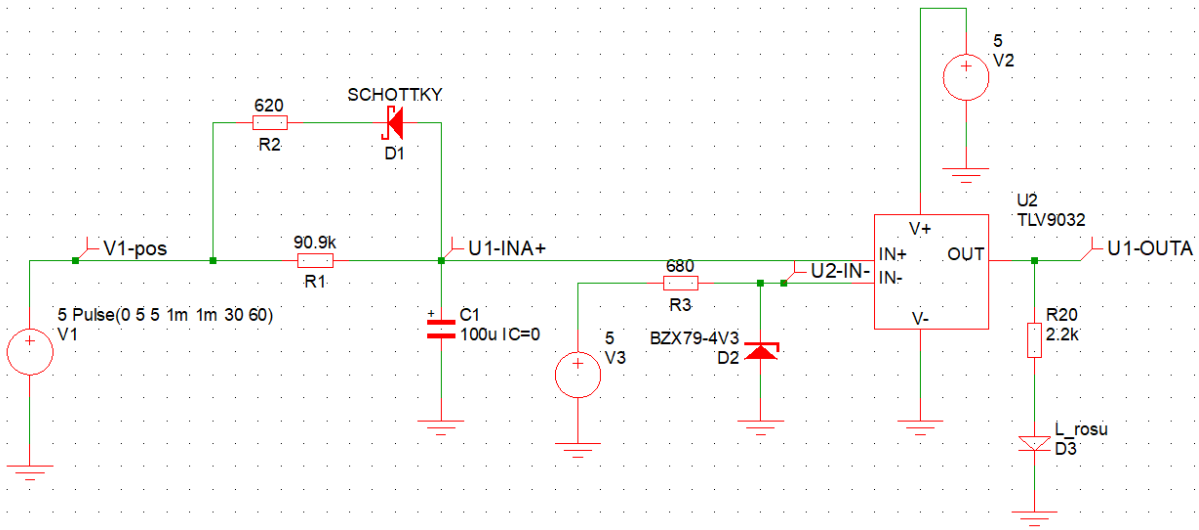
When a pushbutton is pressed, the SET signal is activated, causing the latch to transition to the HIGH state, which then commands the rest of the circuitry.

Later on, it receives the RESET signal, shifting back into the LOW state.



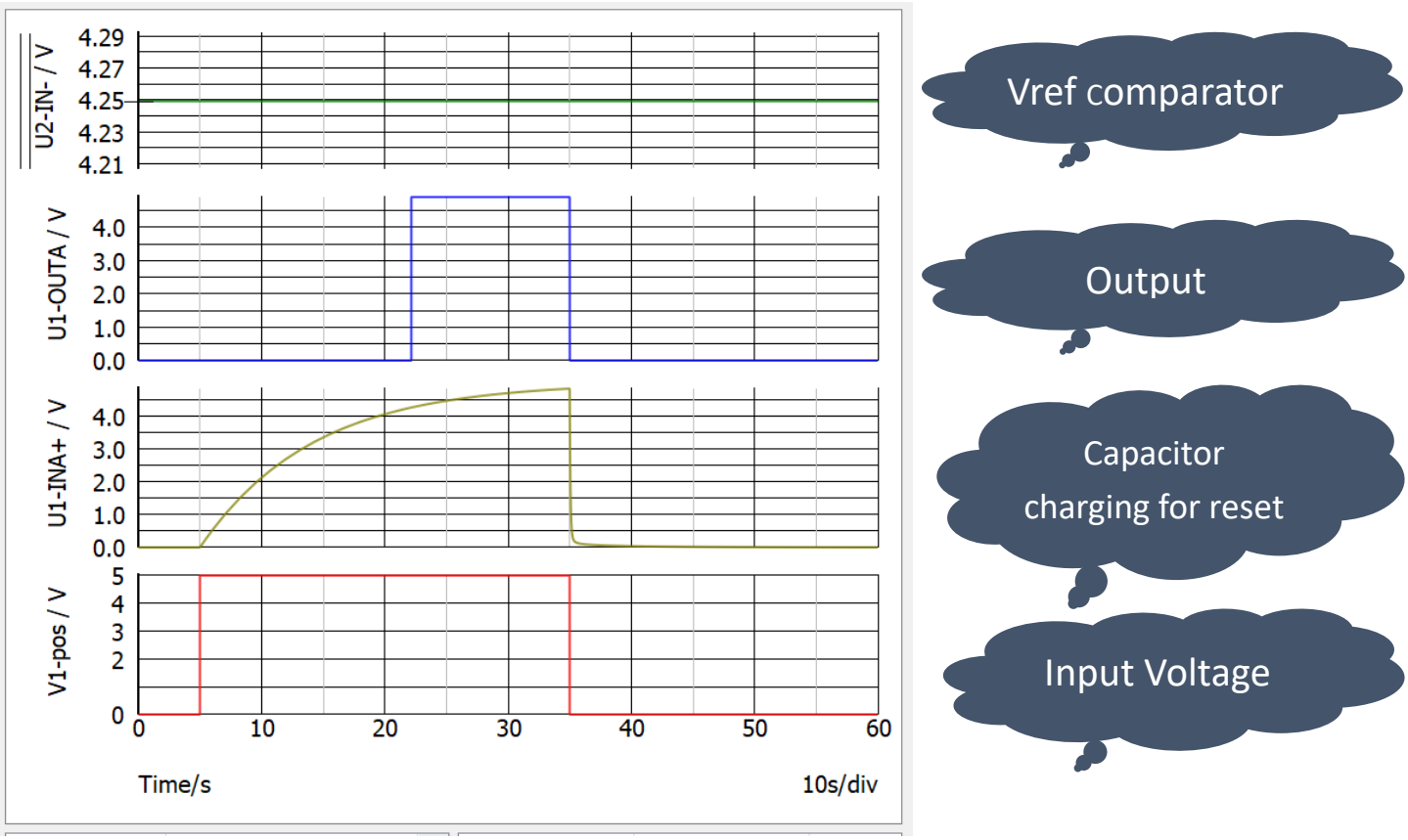
Finally, this is the schematic implementation in Altium of this simple, yet crucial functionality.

1.4. Timed Reset Circuit Simulation & Schematic



The next step was to find a way to prevent this discharge from lasting indefinitely. Preferably, it should end before dropping below the LDO's minimum input voltage.

To achieve this, I utilized the logic of a comparator combined with an RC network.

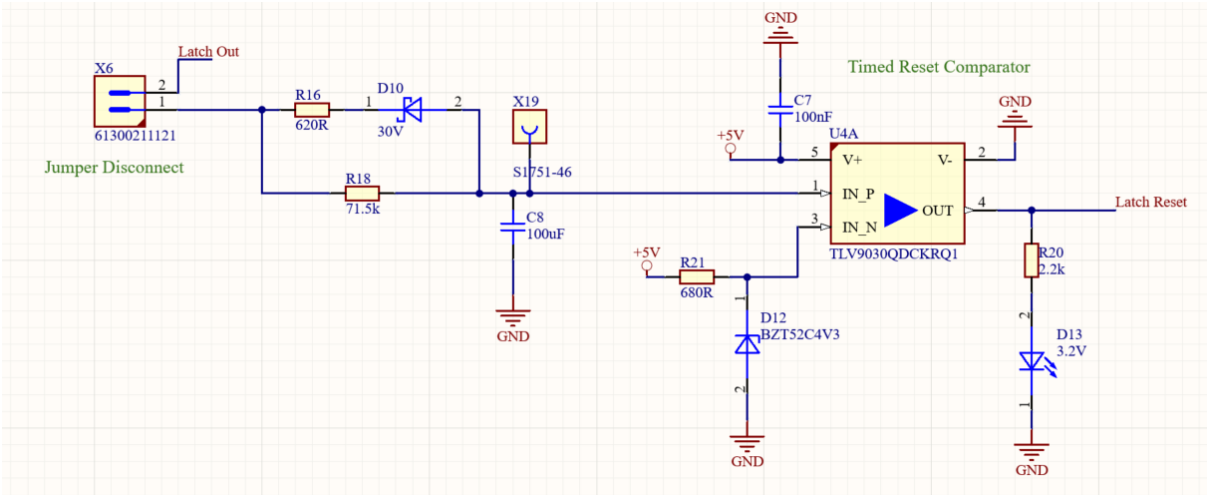


How does it work? The comparator's reference voltage is connected to the inverting input and is set by a Zener diode near the critical threshold of the LDO. Meanwhile, the non-inverting input is controlled by the RC network.

When the latch sends a HIGH signal, the capacitor in the network begins to charge, causing the voltage on the positive terminal to rise. During this time, the main source discharge process is also running.

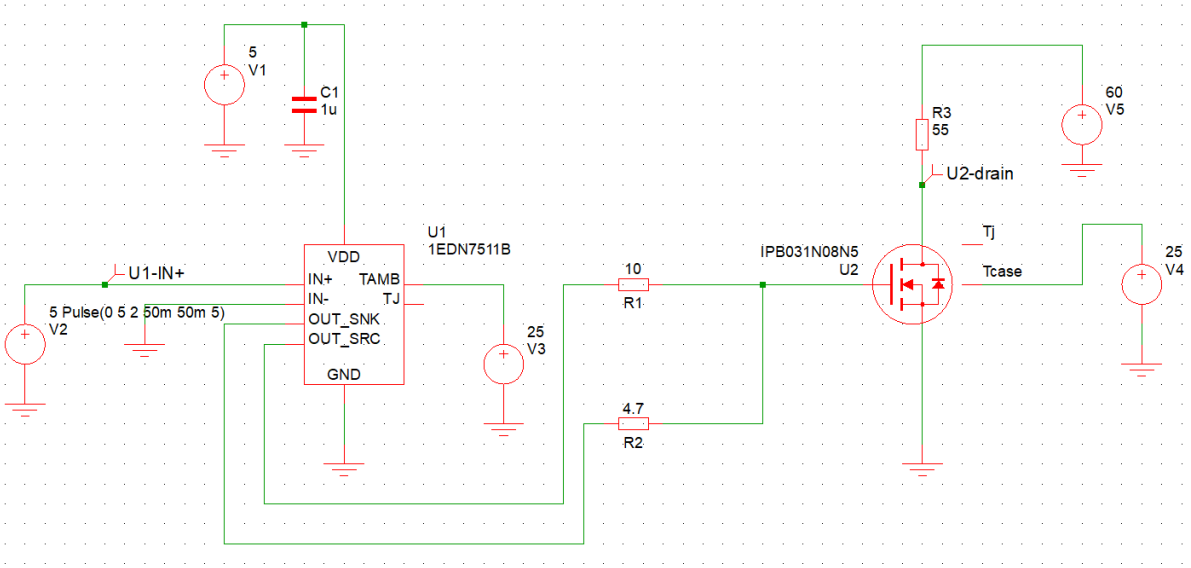
As soon as the RC network voltage reaches the comparator's reference voltage, the comparator switches to HIGH. This sends the RESET signal back to the latch, bringing the entire process to an end.

The RC network was dimensioned so that this takes approximately 15 seconds assuming a constant 60V source.



Here we can see the comparator and the RC network, which act as the active clock of the project, along with the LED signaling for the RESET event.

1.5. Mosfet Low Side Driver Simulation & Schematic



It is now time to introduce the main muscle of the discharge stage, the power transistor along with its dedicated gate driver.



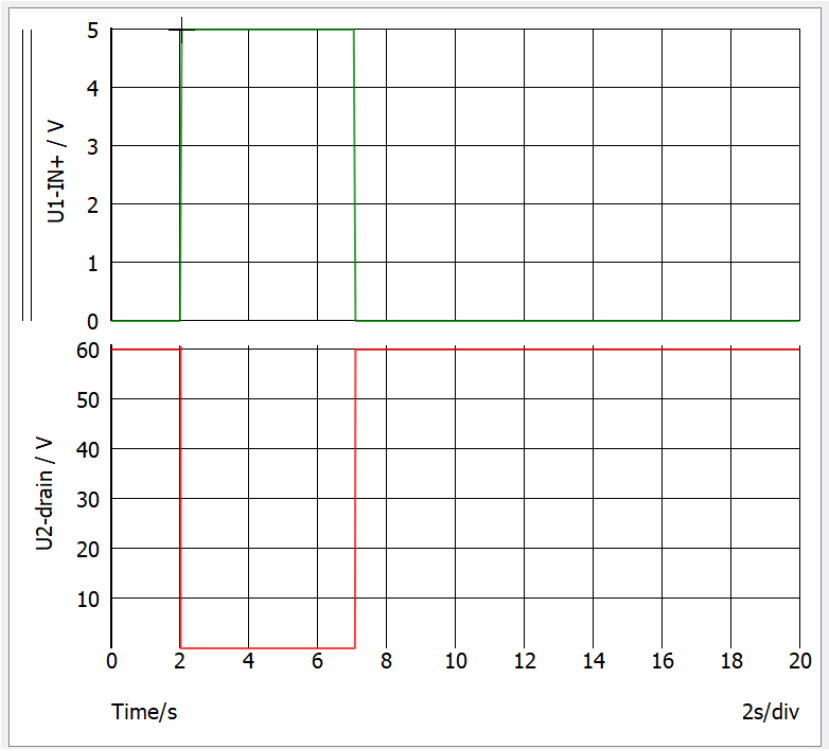
- Supply voltage: 4.5 – 20 V
- Undervoltage lockout: 3.9 – 4.5 V
- Separated source and sink outputs
- 4A source / 8A sink driving capability



- Vds = 80 V, Id = 120 A
- Rds(on)max = 3.1m
- Qg = 69 nC, tr = 18 ns

For this block, I selected these two components from the Infineon portfolio because they perfectly meet my design criteria.

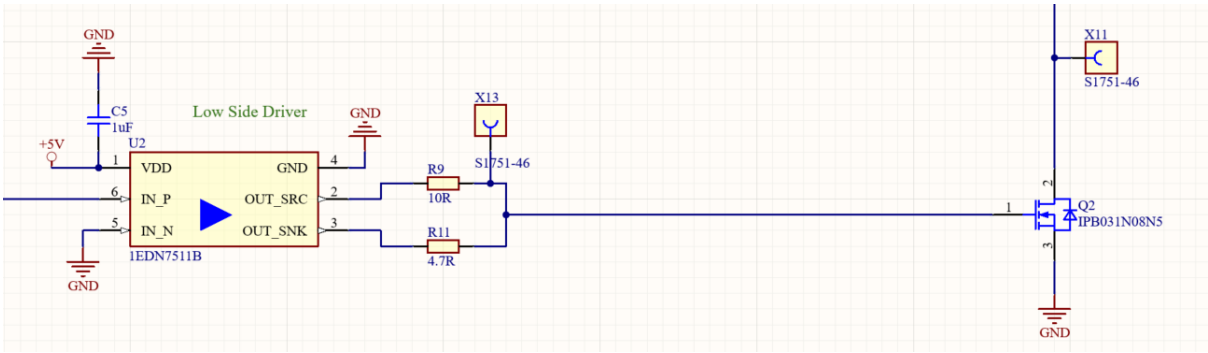
In the case of the gate driver, both its input threshold and its UVLO parameter are significantly below the latch's SET signal, leaving a healthy safety margin. This ensures that the transistor is driven safely and stays well below its maximum operating limits.



Driver

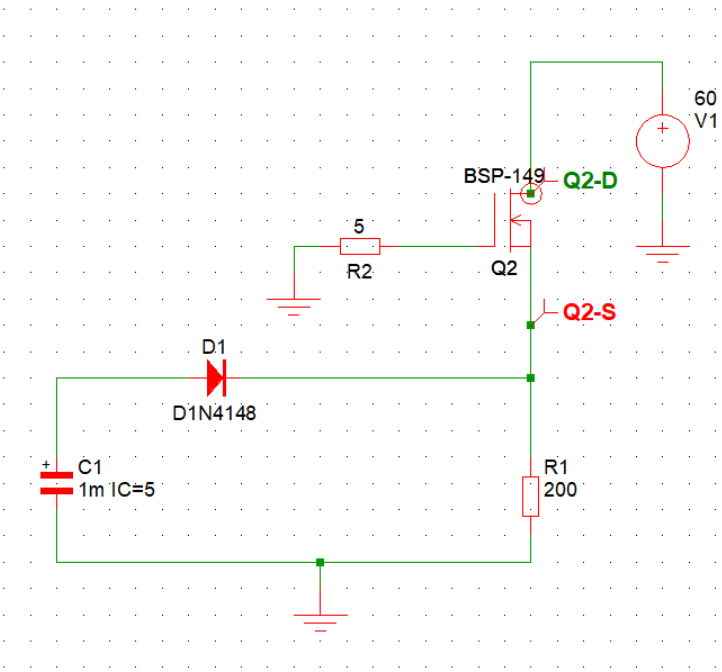
Power Transistor

In this simulation, we can observe that the SET signal coming from the latch and applied to the driver effectively controls the extremely fast switching of the transistor.



Additionally, here is the block as seen in the schematic, highlighting the two main current paths.

1.6. Mosfet Low Side Driver Schematic & Schematic

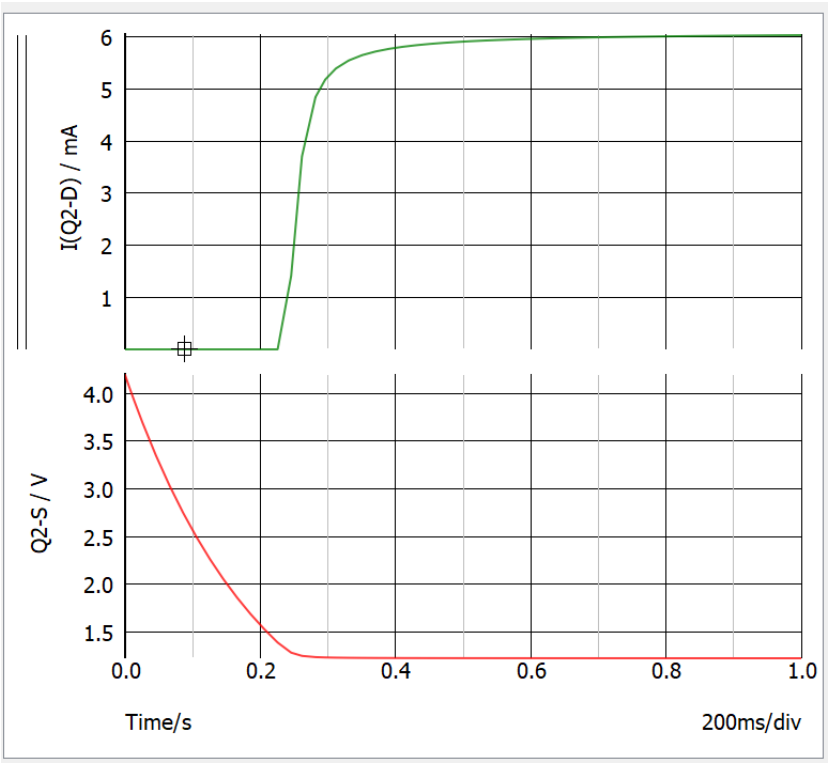


- Ensures fail-safe discharge even if 5V logic power is completely lost.



- Depletion mode
- Vds.max = 200V

Nearing the end, this is where we find the innovation of this project. After the main discharge process stops, the capacitor still holds a residual voltage. This is exactly where the depletion-mode transistor comes into play.

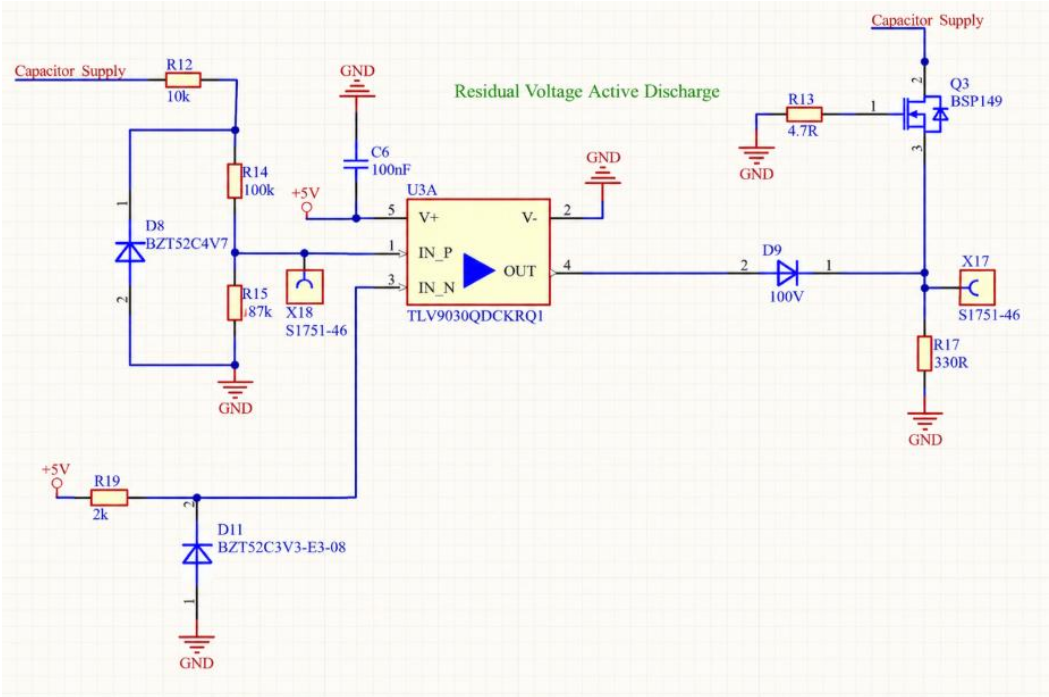


I(Q)

Source Voltage

What does this actually mean? At a strongly negative gate-to-source voltage, the transistor is cut off and does not conduct any current.

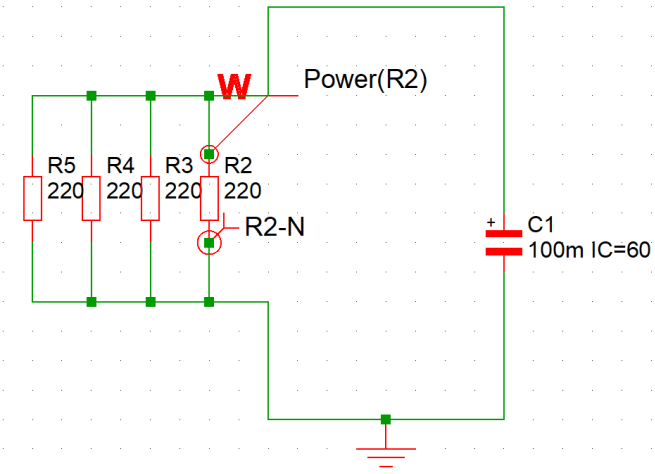
As this voltage approaches zero or a positive value, the transistor opens up and naturally starts conducting current. This forces the remaining voltage left on the capacitor to dissipate through a resistor.



During the main discharge, the comparator is in a HIGH state, applying this voltage to the source of the transistor, which results in a strongly negative gate-to-source potential.

Throughout this time, the transistor remains completely transparent to the circuit. It only opens up shortly after the main discharge process has ended, which is signaled by the comparator switching back to a LOW state.

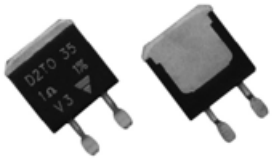
1.7. Power Res Simulation & Schematic





5*R*C=20s
R = 4 / 0.1 = 40 ohm
P = 60^2 / 40 = 90W

Conclusion?
4 resistors in parallel of 220 ohms
Ipeak = 60 / 220 = 0.27 A
Ppeak = 0.27 * 60 = 16.2 W

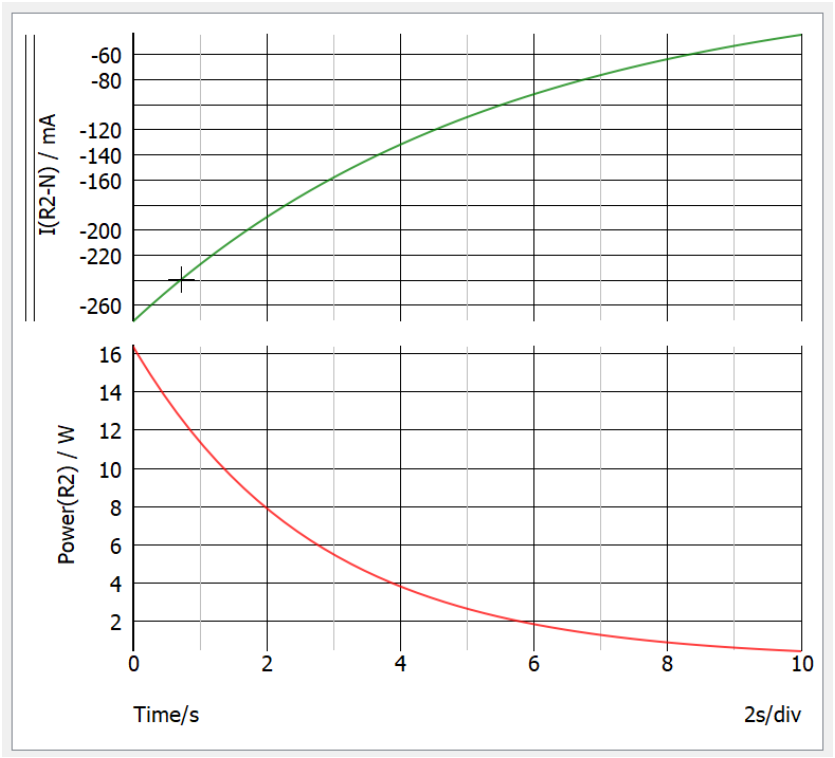


- 35 W
- Voltage rating 500V

And finally, the last block, the power resistors. The principle here is very simple.

Knowing that a capacitor is considered completely discharged after five time constants, and having a strict design constraint for the process to last 20 seconds, the required resistor value was easily calculated.

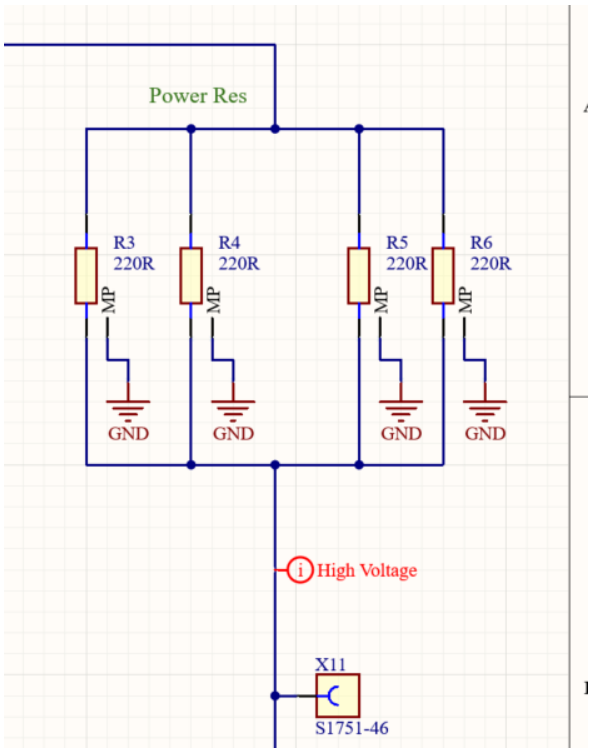
Since a single resistor could not handle the 90W peak power, the trade-off was to use a network of four 220 ohm resistors. This choice ensures that they operate well within their safe operating area, preventing any thermal damage or destruction.



I(R)

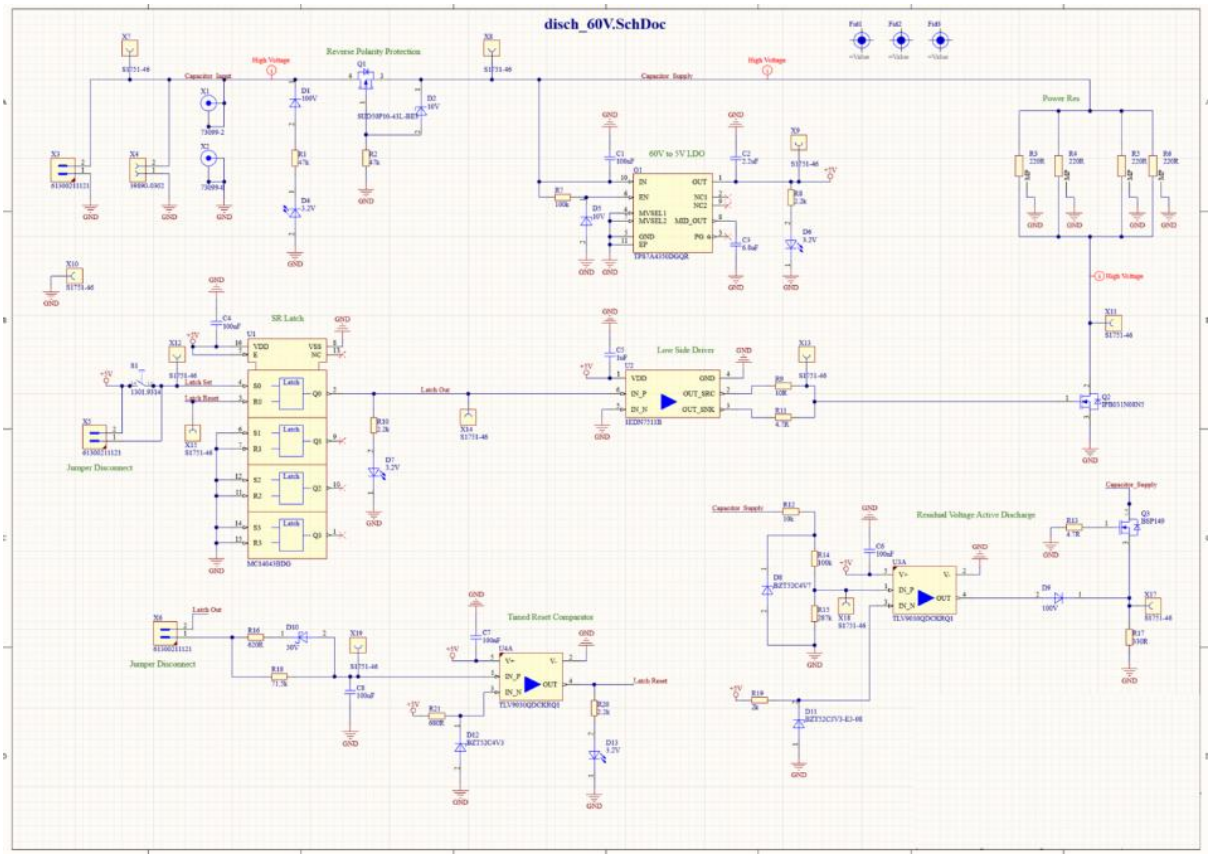
P(R)

The simulation validates my previous calculations, as we can clearly observe the current and power peaks.



And last but not least, here is the Altium implementation, which highlights the third pin connected to GND for efficient energy dissipation.

2. Schematic View Altium



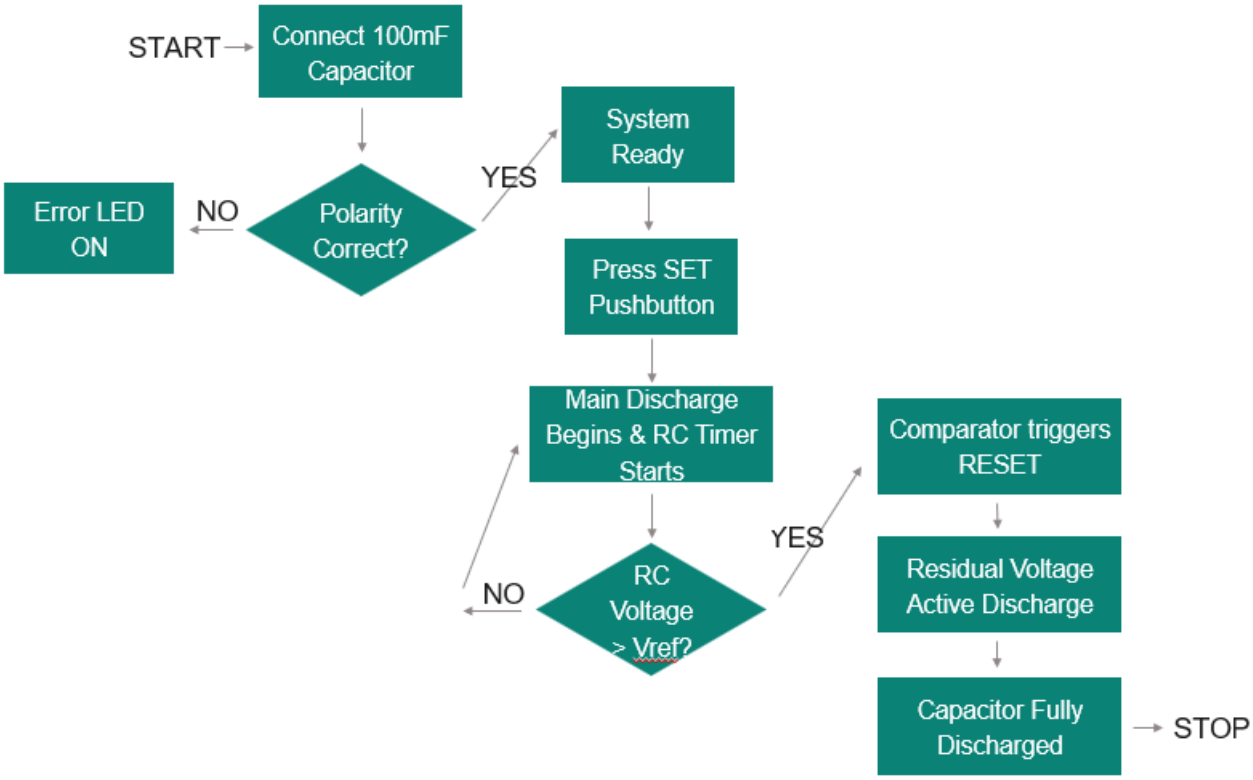
As a summary of everything we have discussed so far, here is the assembly of the entire circuit, operating together as a unified system.

To quickly recap how all these blocks work together, let's look at the system’s flowchart.

Step 1 – I connect the capacitor. If it is connected backwards, the system simply blocks the current and signals an error. If it is correct, the logic board powers up.

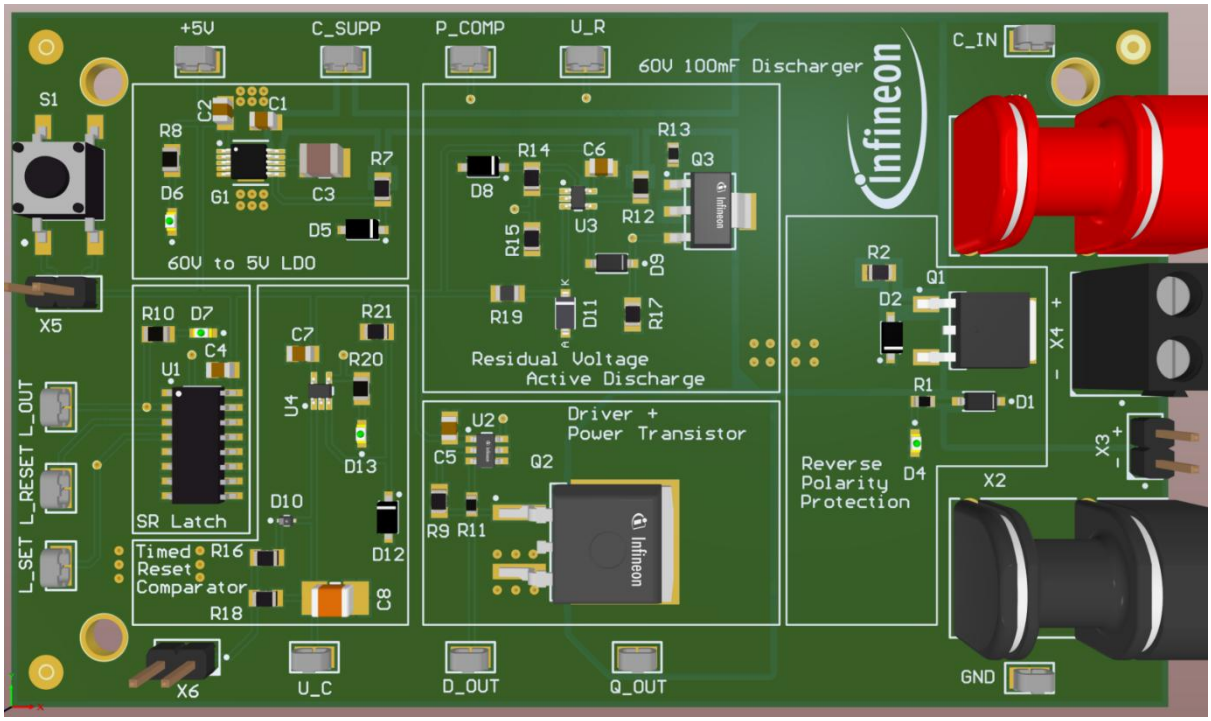
Step 2 – I press the button. This instantly triggers the main discharge and starts our 15 second timer.

Step 3 – Once the timer hits its threshold, it safely stops the main discharge. Finally, the residual circuit automatically takes over to bleed off the remaining voltage, leaving the capacitor completely safe to handle.



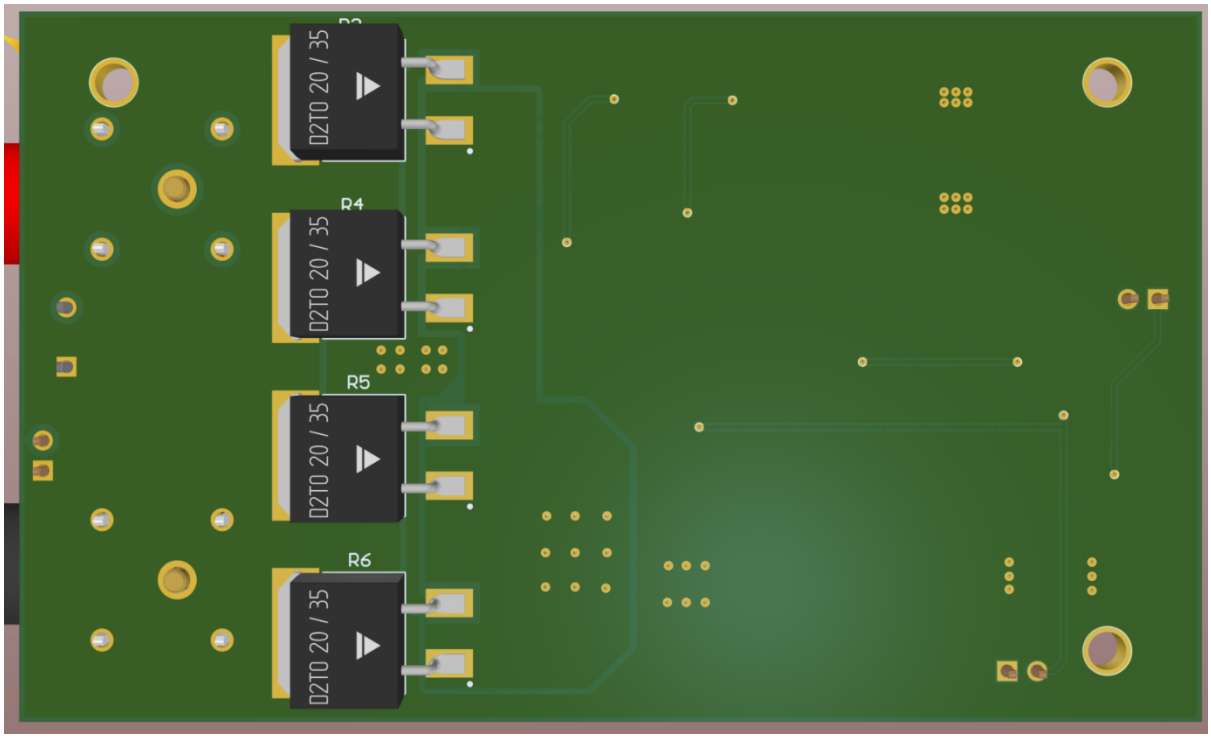
B.Layout Design

1. Overall view 3D – Top View



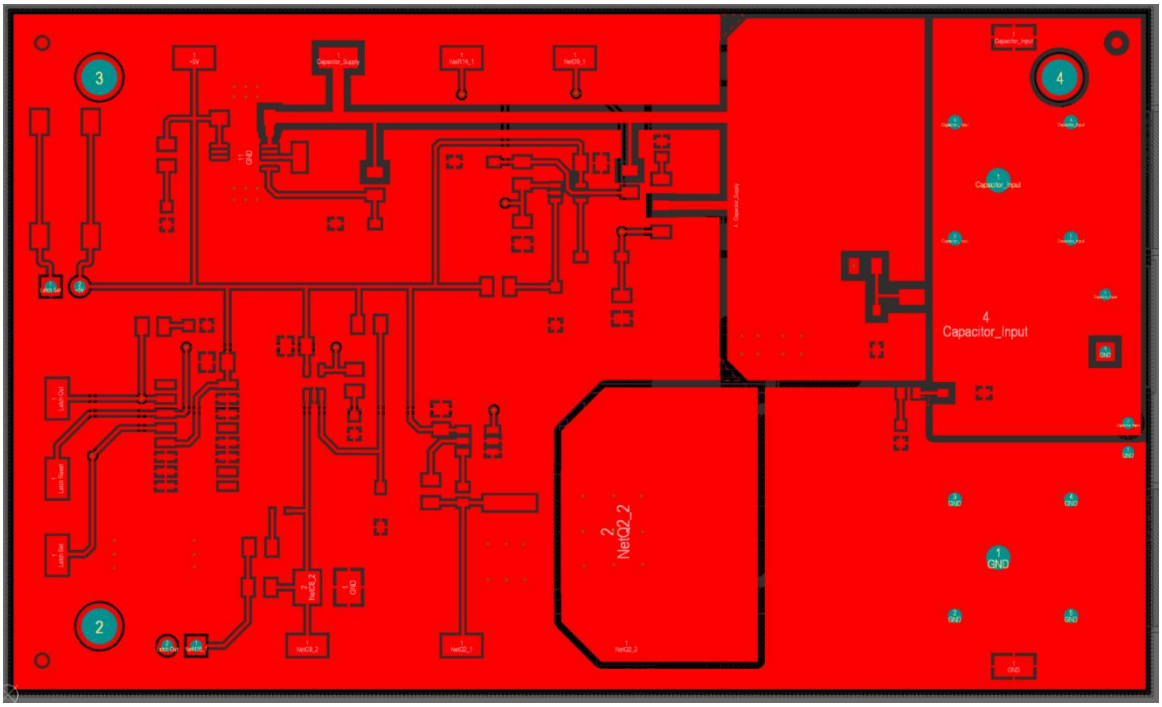
On this picture, we can see how the schematic from a few pages ago has finally come to life. Just like a LEGO game, every component found its place on this tiny board, under a strict size constraint of 100 by 60 millimeters. This specific image captures the top view.

2.Overall view 3D – Bottom View



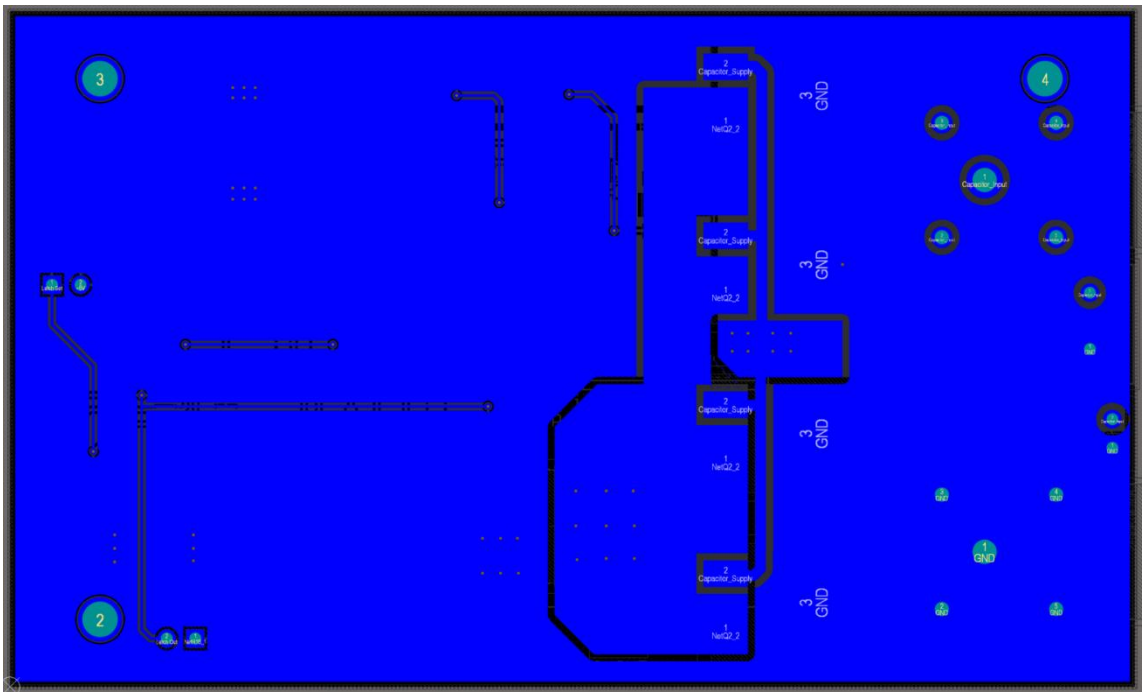
On the bottom view, we can find the four power resistors embedded in a solid ground plane. This plane acts as a thermal heatsink, effectively dissipating all the heat.

3. Overall view 2D – Top View



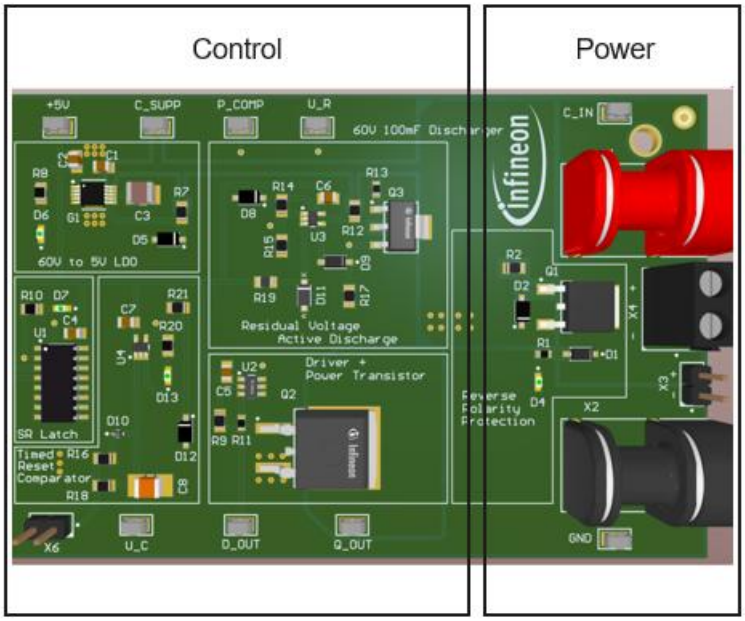
Moving on to the 2D to layer view, we can clearly see the power section, which is made of large copper polygons and thick traces. In contrast, the control and low voltage section consists of short, thin traces. These dimensions were carefully calculated using various design tools according to specs – they are definitely not just arbitrary values.

4. Overall view 2D – Bottom View



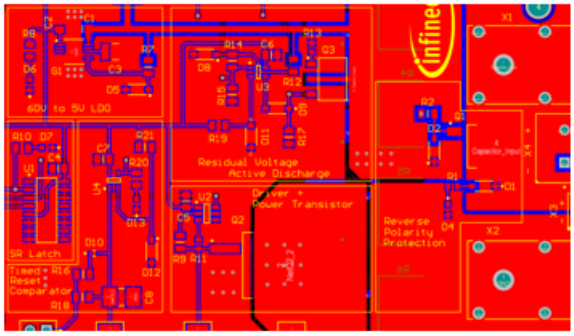
On the 2D bottom layer, you can see the ground plane I mentioned earlier. It serves a dual purpose: it acts as a heatsink and provides the shortest return path for the current, which minimizes current loops and prevents noise pickup.

5. Good Layout Practices

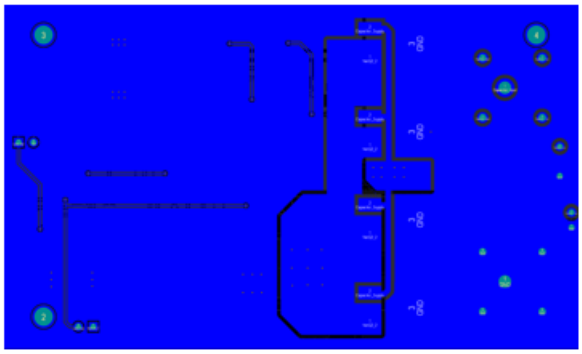


- Functional partitioning – clear boundaries on the silkscreen layer for modularity
- Noise Isolation – sensitive 5V logic placed on the left, high-power 60V switching kept strictly on the right to prevent EMI coupling

Finally, I will review a few design best practices that I followed during this project. Partitioning the layout into functional blocks and isolating sensitive stages ensures not only a clean, visually appealing look, but also guarantees signal integrity and prevents interference.



➢ Thermal management – massive top/bottom copper pours under the power MOSFET and LDO, coupled with via stitching for heat dissipation



➢ GND return path – continuous bottom ground plane to ensure a low-impedance return path for massive discharge currents

Furthermore, implementing solid ground planes on both sides along with via stitching allows for effective thermal management and keeps the current return paths as short as possible.

Intellectual Property Notice

The original Altium Designer project files, including editable schematics, PCB layout files, manufacturing outputs, and the Bill of Materials (BOM), are intentionally excluded from this portfolio due to potential intellectual property and confidentiality restrictions.